



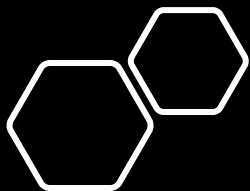
R Packages & Importing data in R

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Recap

- What is R? And why learn R?
- Familiarity with the RStudio interface (opened a file)
- Variable assignment and computations in R
- Basic data types in R
- Data structures in R (Data structures operators)
- Missing values in R



What we will cover today

R Packages

Install and load R Packages

Import files into R

Coding styles and conventions

R Packages

```
... object to mirror_mod.mirror_object
operation == "MIRROR_X":
    mirror_mod.use_x = True
    mirror_mod.use_y = False
    mirror_mod.use_z = False
operation == "MIRROR_Y":
    mirror_mod.use_x = False
    mirror_mod.use_y = True
    mirror_mod.use_z = False
operation == "MIRROR_Z":
    mirror_mod.use_x = False
    mirror_mod.use_y = False
    mirror_mod.use_z = True
```

```
...selection at the end -add
mirror_ob.select= 1
mirror_ob.select=1
context.scene.objects.active
("Selected" + str(modifier_ob
mirror_ob.select = 0
= bpy.context.selected_object
data.objects[one.name].select
print("please select exactly
```

--- OPERATOR CLASSES ---

```
...types.Operator):
    X mirror to the selected
object.mirror_mirror_x"
mirror X"
```

```
context):
context.active_object is not
```

What are R Packages?

- R packages are collections of functions and data sets developed by the community.
- They increase the power of R by improving existing base R functionalities, or by adding new ones.
- Recently, the official repository (CRAN) reached 22,052 packages published.
- A package is a suitable way to organize your own work and, if you want to, share it with others.

Where are R packages stored?

Repository!

- A repository is a place where packages are located so you can install them from it.

Three of the most popular repositories for R packages are:

- CRAN
- Bioconductor
- GitHub

Where are R packages stored?

CRAN: The official repository is a network of FTP and web servers maintained by the R community worldwide. The R Foundation coordinates this process, and for a package to be published here, it must pass several tests that ensure it follows CRAN policies.

Bioconductor: This is a topic-specific repository intended for, intended for open-source software for bioinformatics.

GitHub: GitHub is probably the most popular repository for open-source projects.

How to install an R package: From CRAN

There are two methods:

1. From the RStudio interface
2. Use the command **install.packages("package_name")**

Load/Unload a package in R

- To load a package: Do this with the **library(package_name)** command
- To unload a package: Do this with the **detach("package:readxl", unload = TRUE)**

P.S.: It is possible to load a package with **require(package_name)**, but the difference is that it will not throw an error if the package is not installed.

Update, Remove and Check Installed Packages

- To check what packages are installed on your computer, you can use:
installed.packages()
- Uninstall a package with the function: **remove.packages("packagename")**
- You can check what packages need an update with a call to the function:
old.packages()
- You can update all packages by using: **update.packages()**
- Use the RStudio interface

Importing Data into R

- Import a space separated .txt file
- Import a comma separated .txt file
- Import a .csv file
- Import a .xlsx file
- Import a .dta file
- Import a .sav/.spss file

Suggestions on Coding Style

1. First things first. Use RStudio. It's a great IDE for R. It saves you a lot of time and effort.
2. R is case-sensitive – be careful!
3. Use a naming convention. It will help you and others to read and understand your code. You should be consistent throughout your code. You can read this [article](#) for more information.
4. Use coding standards. Similar to naming convention, it will help you and others to read and understand your code. [Hadley Wickham's](#) or [Colin Gillespie's](#) guides are very helpful.
5. Finally, document your codes and functions. **Make sure to comment your codes.**

Resources

- There are lots of free online resources to learn R
 - better to find the ones that fit your needs
- 1. R and RStudio online learning: hit the home button in the Help pane
 - Cheat sheets
 - Books
- 2. Software Carpentry has great free lessons on R programming
 - <http://swcarpentry.github.io/r-novice-inflammation/>
 - <https://swcarpentry.github.io/r-novice-gapminder/>
- 3. Books:
 - [The R Book](#)
 - [R for Data Science](#) (online open source)



Other Resources

- <https://swcarpentry.github.io/r-novice-inflammation/13-supp-data-structures.html>
- <https://www.geeksforgeeks.org/data-structures-in-r-programming/>
- <https://data-flair.training/blogs/data-structures-in-r/>
- https://warwick.ac.uk/fac/sci/moac/degrees/moac/ch923/r_introduction/help/
- <https://www.r-bloggers.com/2013/01/how-to-install-packages-on-r-screenshots/>
- <https://cran.r-project.org/web/packages/policies.html>
- **Important:** <https://statsandr.com/blog/descriptive-statistics-in-r/>

Thank You

